

Linear Algebra Notes

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1. What is Linear Algebra?

We will not answer this question immediately, except to say that linear algebra is the study of *linear transformations*. Here are some definitions and axioms to remember:

Definitions

1. A vector is an element of a vector space
2. Vectors are denoted by small bold letters, say $\vec{v}$ and matrices by capitals, A .
3. An $m \times n$ matrix means m rows and n columns. We will deal with finite matrices.
4. Notation: $A = (a_{j,k})_{j=1,k=1}^{m,n}$. The entries of the matrix A will be denoted $[A]_{ij}$ or a_{ij} .

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

5. The transpose is defined by swapping rows and columns. $[A^T]_{ij} = [A]_{ji}$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

2. Axioms

Field Axioms

Let $\mathbb{F}$ be a non-empty set. Let $+$ and $\times$ be two functions (operations) from $\mathbb{F} \rightarrow \mathbb{F}$. $\mathbb{F}$ satisfies

1. $a + b = b + a$
2. $(a + b) + c = a + (b + c)$ $(ab)\vec{v} = a(b\vec{v})$
3. There exists 0 such that $a + 0 = a$
4. There exists b such that $a + b = 0$
5. $(ab)c = a(bc)$
6. There exists 1 such that $1 \times a = a \times 1 = a$
7. There exists inverse such that $a^{\{-1\}} \times a = 1$
8. $ab = ba$
9. $a(b + c) = ab + ac$

Vector Spaces

Let $\mathbb{V}$ be a non-empty set. Let $\vec{v}$, $\vec{w}$ and $\vec{u}$ be arbitrary elements (called *vectors*, represented with a bold typeface and arrow on top) of $\mathbb{V}$, and a and b be scalars belonging to $\mathbb{F}$. $\mathbb{V}$ is called a *vector space* if it satisfies the following:

1. $\vec{v} + \vec{w} = \vec{w} + \vec{v}$.
2. $\vec{v} + (\vec{w} + \vec{u}) = (\vec{v} + \vec{w}) + \vec{u}$
 $(ab)\vec{v} = a(b\vec{v})$
3. There exists $\vec{0}$ such that $\vec{v} + \vec{0} = \vec{v}$
4. There exists $\vec{w}$ such that $\vec{v} + \vec{w} = \vec{0}$
5. $1\vec{v} = \vec{v}$
6. $a(\vec{v} + \vec{w}) = a\vec{v} + a\vec{w}$
 $(a + b)(\vec{v}) = a\vec{v} + b\vec{v}$

Note that the operations together with the set make the field. For instance $(\mathbb{R}, +, \times)$ is a field. $(\mathbb{R}, +, *)$ where $a * b = 2ab$ is also a field (check this!). Another more familiar example may be the integers modulo a number, $\mathbb{Z}/n\mathbb{Z}$. Usually we will use the regular multiplication and addition. The elements of a field are called *scalars*

I will sometimes forget the $\mathbb{V}$ and the $\mathbb{F}$. We call $\mathbb{V}$ a vector space over $\mathbb{F}$.

Properties

Claim. $\vec{0}$ is unique.

Proof. Suppose there are two zeros, $\vec{0}_1$ and $\vec{0}_2$. Then

$$\vec{0}_2 = \vec{0}_2 + \vec{0}_1 = \vec{0}_1 + \vec{0}_2 = \vec{0}_1.$$

Claim. $\vec{0}$ is unique.

Proof. Suppose there are two zeros, $\vec{0}_1$ and $\vec{0}_2$. Then

$$\vec{0}_2 = \vec{0}_2 + \vec{0}_1 = \vec{0}_1 + \vec{0}_2 = \vec{0}_1.$$

Claim. Additive inverses are unique.

Proof. Suppose there are two inverses to $\vec{v}$, $\vec{w}_1$ and $\vec{w}_2$. Then

$$\vec{v} + \vec{w}_1 = \vec{v} + \vec{w}_2$$

$$\vec{v} + \vec{w}_1 + \vec{w}_1 = \vec{v} + \vec{w}_2 + \vec{w}_1$$

$$\vec{w}_1 = \vec{w}_2.$$

Claim. $\vec{0} = 0\vec{v}$

Proof.

$$0(\vec{v} + \vec{v}) = 0(\vec{v}) + 0(\vec{v})$$

$$(2)(0)\vec{v} = 0\vec{v} = 0\vec{v} + 0\vec{v}$$

Adding the additive inverse (say $\vec{w}$) on both sides

$$0\vec{v} + w = 0\vec{v} + 0\vec{v} + w$$

$$\vec{0} = 0\vec{v}$$

Claim. $\vec{v} + (-1)\vec{v} = 0$

$$\vec{v} + (-1)\vec{v} = 1v + (-1)\vec{v} = \vec{v}(1 - 1) = 0\vec{v} = \vec{0}$$

Linear Transformation

Let $\mathbb{V}$ and $\mathbb{W}$ be vector spaces. A transformation $T : \mathbb{V} \rightarrow \mathbb{W}$ is called *linear* if:

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$
2. $T(a\vec{v}) = aT(\vec{v})$ for all $\vec{v} \in \mathbb{V}$ and all scalars $a \in \mathbb{F}$.

Basis, Span and Linear Independence

3. Matrix Multiplication

3.1. Matrix-Vector

3.2. Matrix

3.3. Inverses

Inverses

Left Invertible: A is invertible if there exists B such that $BA = I$

Right Invertible: A is invertible if there exists C such that $AC = I$

Example:

$$(0.5 \ 0.5) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Invertible: A is invertible if both B and C exist.

Claim. If a linear transformation is invertible, then its left and right inverses coincide and are unique.

Proof.

$$B_1 = B_1 I = B_1 (AB) = (B_1 A) B = IB = B$$

$$B = BI = B(AC) = (BA)C = IC = C$$

Claim. An invertible matrix is square.

4. Subspaces

Subspaces

A *subspace* of a vector space $\mathbb{V}$ is a non-empty subset of it that is closed under multiplication and addition. $W \subset \mathbb{V}$ is a subspace if

- $\vec{w}_1 + \vec{w}_2 \in W$ for all $\vec{w}_1, \vec{w}_2 \in W$
- $c\vec{w} \in W$ for all $\vec{w} \in W$ and $c \in \mathbb{F}$

Another way to say this is all linear combinations are contained in the subspace. A corollary is that $\vec{0} \in W$.

If X and Y are two subspaces, $X \cap Y$ is also a subspace.

$$X + Y \stackrel{\text{def}}{=} \{ \vec{v} \mid \vec{v} = \vec{x} + \vec{y}, x \in X, y \in Y \}$$

Some Subspaces

Let A be a linear transformation.

The *nullspace* or *kernel*, denoted $\text{Null}(A)$ or $\text{Ker}(A)$ is the set of all vectors that satisfy $A\vec{x} = \vec{0}$.

The *column space* or *range*, denoted $\text{Col}(A)$ or $\text{Ran}(A)$ is the set of all linear combinations of the columns of A , that is, the set of all vectors $\{\vec{w} \mid \vec{w} = A\vec{v}, \vec{v} \in A\}$.

The span of a set of vector $L = \{\vec{v}_1, \vec{v}_2, \dots\}$ is the set of all its linear combinations. That is,

$$\text{span}(L) = \left\{ \sum_i a_i \vec{v}_i \mid a_i \in \mathbb{F}, \vec{v}_i \in L \right\}$$

Exercise: Prove that these are subspaces.

5. Solving Linear Systems

$$\begin{array}{cccccccl} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

We will write this in matrix notation.

$$A\vec{x} = \vec{b}$$

$$A = \underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}}_{\text{Coefficient Matrix}}$$

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & | & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & | & b_2 \\ \vdots & \vdots & \ddots & \vdots & | & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & | & b_m \end{pmatrix}}_{\text{Augmented Matrix}}$$